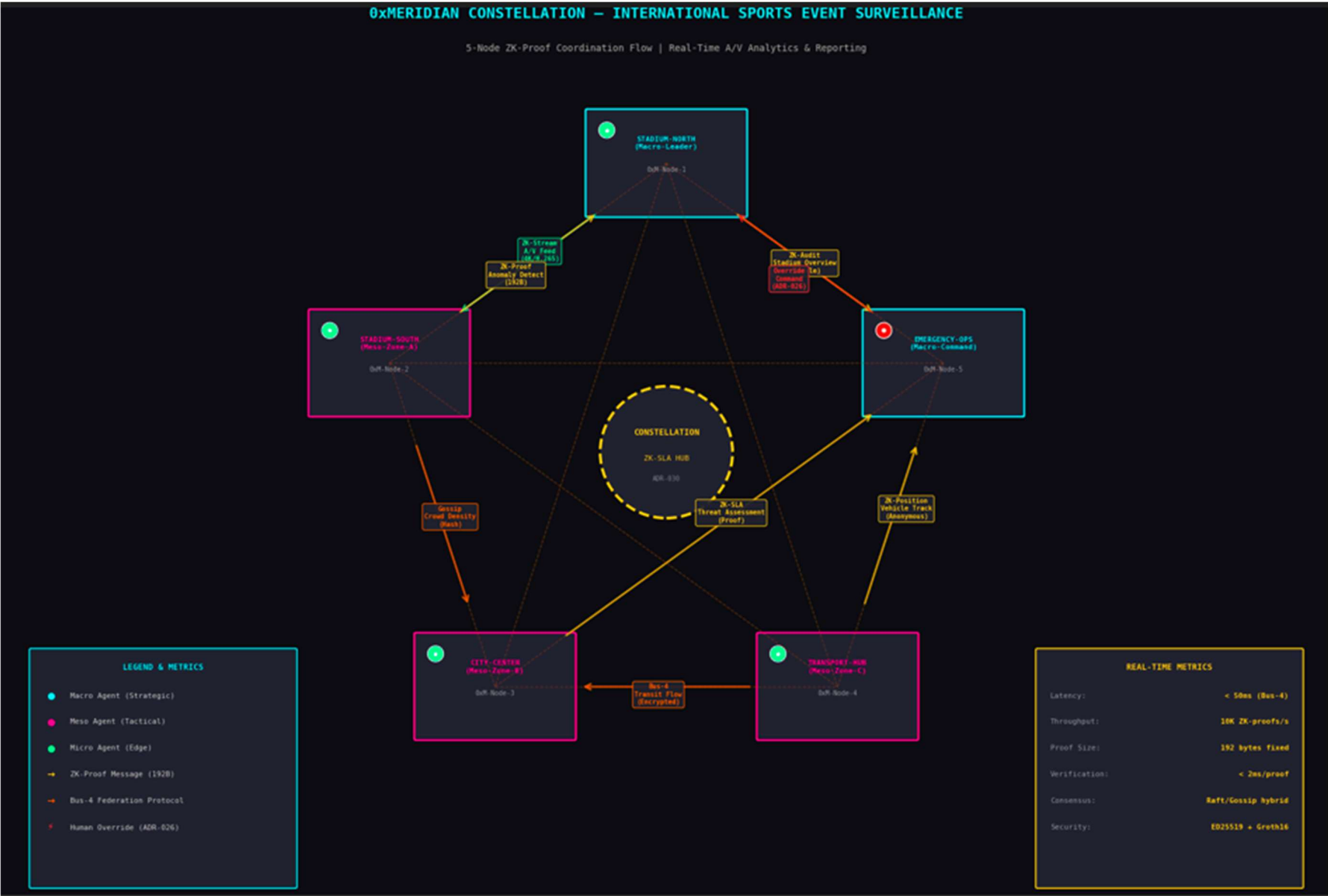


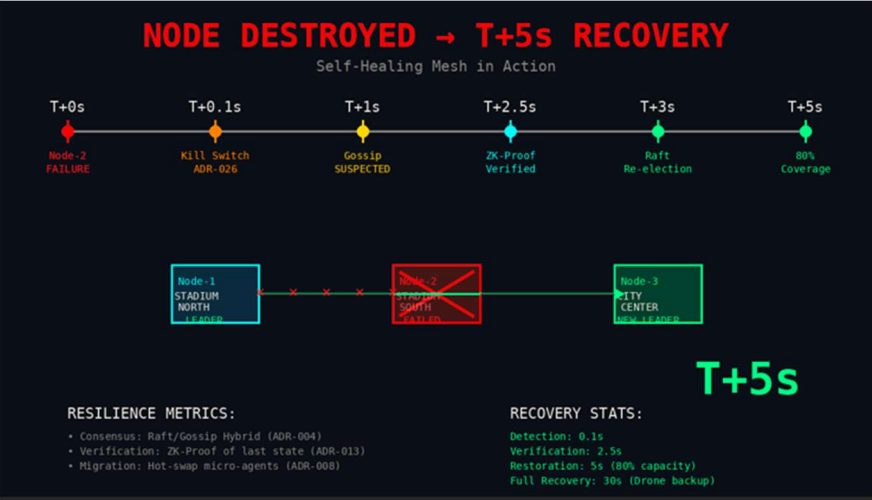
0xMERIDIAN CONSTELLATION: Next-Gen Surveillance for Major Media Events: AI at the Edge



🏠 CONSTELLATION ARCHITECTURE

Node	Role	Tier	Hardware	Responsibilities
0xM-Node-1	STADIUM-NORTH	Macro-Leader	Mac Studio Ultra (128GB)	North coordination, multi-camera 4K video
0xM-Node-2	STADIUM-SOUTH	Meso-Zone-A	MacBook Pro M3 Max	Crowd analysis, anomaly detection
0xM-Node-3	CITY-CENTER	Meso-Zone-B	Mac Mini Cluster	Public transport monitoring
0xM-Node-4	TRANSPORT-HUB	Meso-Zone-C	Edge Server ARM64	Vehicle flow tracking
0xM-Node-5	EMERGENCY-OPS	Macro-Command	Secure Workstation	Emergency command, human override

🚨 DISASTER RECOVERY SCENARIOS



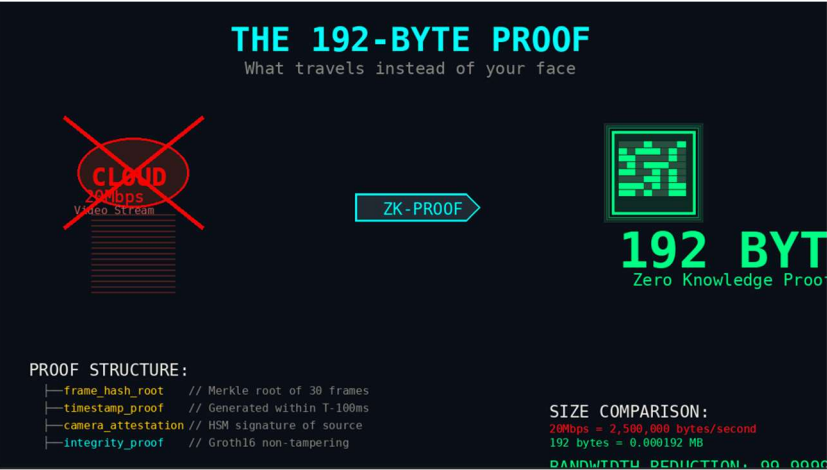
Scenario A: Physical Node Compromise

Timestamp	Event
T+0s	Node-2 detects physical intrusion
T+0.1s	ADR-026 Kill Switch activated locally
T+0.5s	Gossip SWIM marks Node-2 as SUSPECTED
T+1s	Ambassador Agents (Node-1, Node-3, Node-5) verify ZK-Proof of Node-2 last state
T+2s	Node-2 declared FAILED
T+2.5s	Raft re-election: Node-3 becomes South Zone leader
T+3s	Micro-agents of Node-2 migrated to Node-3 (hot-swap via ADR-008)
T+5s	South Zone coverage restored to 80%
T+30s	Backup Drone with embedded 0xMeridian arrives in position → full recovery

Scenario B: Network Partitioning (Split-Brain)

(Details covered in Section 6: Recovery & Reconciliation)

 ZK-PROOF MESSAGING FLOWS



SECTION 1: VIDEO STREAMS & ANALYSIS (Node-1 ↔ Node-2)

Step	Timestamp	Origin	Destination	Message Type	Protocol/Payload	Size	Latency	ADR	ZK-Proof	Action
1.0	T+0ms	Camera IP-4K (External)	Node-1 STADIUM-NORTH	Raw A/V Stream	H.265 encoded, 4K@30fps, 20Mbps	~2.5MB/s	<5ms	ADR-001	—	Frame acquisition
1.1	T+15ms	Node-1 Interface Agent	Node-1 Memory Agent	Frame Parse	Decoded tensor (224×224×3), metadata	150KB	15ms	ADR-001, ADR-005	—	Input normalization

Step	Timestamp	Origin	Destination	Message Type	Protocol/Payload	Size	Latency	ADR	ZK-Proof	Action
1.2	T+95ms	Node-1 Memory Agent	Node-1 Planner Agent	RAG Context	FAISS retrieval: historical threat profiles (top-K=5)	80KB	80ms	ADR-005, ADR-006	—	Context augmentation
1.3	T+275ms	Node-1 Planner Agent	Node-1 Magellano Engine	DAG Execution	Inference request: "analyze_threat_sector_7A"	2KB	180ms	ADR-002, ADR-004	—	Task graph generation
1.4	T+375ms	Node-1 Magellano 3.3B	Node-1 Critic Agent	Inference Result	Threat level: 0.87, Bounding boxes, Class labels	50KB	100ms	ADR-003	—	Analysis completed
1.5	T+495ms	Node-1 Critic Agent	Node-1 State Manager	Quality Validation	Score: 0.91 (≥ 0.85 threshold), Approved	1KB	120ms	ADR-008	—	Output validation
1.6	T+500ms	Node-1 ZK-Prover	Node-1 Bus-4 Interface	ZK-Stream Init	Stream attestation: "Frames 184700-184729 valid"	192B	100ms	ADR-013	Groth16 proof	Integrity proof generation
1.7	T+600ms	Node-1	Node-2 STADIUM-SOUTH	ZK-Stream A/V	H.265 + ZK-Proof + Merkle root frame	2.5MB + 192B	<50ms	ADR-006, ADR-012	Included	Secure transmission

Step	Timestamp	Origin	Destination	Message Type	Protocol/Payload	Size	Latency	ADR	ZK-Proof	Action
1.8	T+602ms	Node-2 ZK-Verifier	Node-2 Memory Agent	Verification	Verify Groth16: TRUE, Merkle match: TRUE	192B	2ms	ADR-013	Verified	Stream acceptance
1.9	T+650ms	Node-2 Magellano 77M	Node-2 Critic Agent	Cross-Validation	Crowd density analysis: 4.2 ppl/m²	30KB	200ms	ADR-003	—	Secondary analysis

SECTION 2: GOSSIP & ZONE COORDINATION (Node-2 ↔ Node-3)

Step	Timestamp	Origin	Destination	Message Type	Protocol/Payload	Size	Latency	ADR	ZK-Proof	Action
2.0	T+850ms	Node-2	Node-1	ZK-Proof Anomaly	"Anomaly detected: RAPID_CROWD_CONVERGENCE"	192B	<50ms	ADR-013	Groth16 proof	Validation feedback
2.1	T+200ms	Node-2 Gossip Agent	Node-3 Gossip Agent	SWIM Ping	Health check: Node-2 ACTIVE, load: 0.67	64B	200ms	ADR-004	—	Membership protocol
2.2	T+400ms	Node-2	Node-3	Gossip Density Update	Zone density map: {7A: 4.2, 7B: 3.1, 8A: 2.8}	500B	200ms	ADR-004	Hash digest	State propagation

Step	Timestamp	Origin	Destination	Message Type	Protocol/Payload	Size	Latency	ADR	ZK-Proof	Action
2.3	T+600ms	Node-3	Node-2	Gossip Ack	Ack sequence 184729, Merkle root sync	128B	200ms	ADR-004	—	Receipt confirmation
2.4	T+800ms	Node-3 Planner Agent	Node-3 Ambassador	Aggregation Task	"Correlate crowd data from Node-2 + local transit"	5KB	180ms	ADR-002, ADR-030	—	Fusion planning
2.5	T+1000ms	Node-3 Magellano 3.3B	Node-3 ZK-Prover	Threat Assessment	Global threat: 0.89 (Node-2: 0.87 + Node-3: 0.65)	40KB	100ms	ADR-003	—	Aggregated inference
2.6	T+1100ms	Node-3 ZK-Prover	Node-3 Bus-4 Interface	ZK-SLA Generation	"Aggregation correct, sources: 2, confidence: 0.89"	192B	100ms	ADR-013, ADR-030	Groth16 proof	Proof of correct aggregation

SECTION 3: ESCALATION & STRATEGIC COMMAND (Node-3 → Node-5)

Step	Timestamp	Origin	Destination	Message Type	Protocol/Payload	Size	Latency	ADR	ZK-Proof	Action
3.1	T+1200ms	Node-3 Ambassador	Node-5 Ambassador	ZK-SLA Threat Assessment	Service: CROWD_SAFETY_ASSESSMENT, Threat: 0.89	192B + metadata	<50ms	ADR-030	Groth16 proof	Send assessment
3.2	T+1202ms	Node-5 ZK-Verifier	Node-5 Critic Agent	Verification	Verify: TRUE, Source reputation: 0.97, Credits: 50	192B	2ms	ADR-013	Verified	SLA Validation
3.3	T+1250ms	Node-5 Critic Agent	Node-5 Macro Planner	Escalation Eval	"Threat 0.89 > 0.85, auto-escalate? NO (human policy)"	2KB	120ms	ADR-008	—	Escalation decision
3.4	T+1300ms	Node-5	Node-1, Node-2, Node-3, Node-4	Status Query	"Request consolidated situational awareness"	1KB	50ms	ADR-006	—	Global state request
3.5	T+1350ms	Node-1	Node-5	ZK-Audit Merkle	Merkle root: 0xABC123..., 3000 events, 100ms batch	64B + proof 192B	<50ms	ADR-011, ADR-013	Merkle + ZK	North audit trail
3.6	T+1400ms	Node-2	Node-5	ZK-Audit Merkle	Merkle root: 0xDEF456..., 2800 events	64B + proof 192B	<50ms	ADR-011, ADR-013	Merkle + ZK	South audit trail

Step	Timestamp	Origin	Destination	Message Type	Protocol/Payload	Size	Latency	ADR	ZK-Proof	Action
3.7	T+1450ms	Node-4	Node-5	ZK-Position Vehicle Track	"Vehicle XYZ in zone, speed 45km/h, route: Stadium"	192B	<50ms	ADR-013	ZK-Position	Anonymous tracking
3.8	T+1500ms	Node-5 Macro Planner	Node-5 Decision Engine	Situational Fusion	"Multi-source threat confirmed, recommend action"	10KB	200ms	ADR-002	—	Decision fusion

SECTION 4: TRANSPORT & MOBILITY (Node-4 ↔ Node-3)

Step	Timestamp	Origin	Destination	Message Type	Protocol/Payload	Size	Latency	ADR	ZK-Proof	Action
4.1	T+0ms (continuous)	ANPR Camera Node-4	Node-4 Micro Agent	Vehicle Detection	Plate hash, timestamp, direction vector	256B	<10ms	ADR-002	—	Edge detection
4.2	T+50ms	Node-4 Micro Agent	Node-4 Meso Agent	Flow Aggregation	10 vehicles/min, route analysis, congestion: 0.3	1KB	40ms	ADR-002	—	Local aggregation
4.3	T+200ms	Node-4 Magellano 77M	Node-4 ZK-Prover	Route Prediction	"Predicted arrival: Stadium North, T+15min"	2KB	200ms	ADR-003	—	Flow prediction
4.4	T+400ms	Node-4	Node-3	Bus-4 Transit Flow	Encrypted: vehicle flows, predictions, anomalies	5KB	<50ms	ADR-006, ADR-012	Encrypted	Data sharing

Step	Timestamp	Origin	Destination	Message Type	Protocol/Payload	Size	Latency	ADR	ZK-Proof	Action
4.5	T+450ms	Node-3	Node-4	Flow Ack + Correlation	"Correlated with crowd density, risk: MODERATE"	2KB	<50ms	ADR-006	—	Integrated feedback

SECTION 5: HUMAN OVERRIDE & CRITICAL COMMAND (Node-5 → All)

Step	Timestamp	Origin	Destination	Message Type	Protocol/Payload	Size	Latency	ADR	ZK-Proof	Action
5.1	T+2000ms (trigger)	Human Operator (HSM)	Node-5 Override Interface	Emergency Command	"INITIATE_EVACUATION_PROCEDURE" + biometric auth	2KB	<100ms	ADR-026	Ed25519 + ZK	Cryptographic override
5.2	T+2100ms	Node-5 Override Verifier	Node-5 Kernel	Auth Validation	HSM signature: VALID, ZK-Proof: VALID, Level: 5	1KB	50ms	ADR-026	Verified	Override validation
5.3	T+2150ms	Node-5	Node-1	OVERRIDE Command	"EVACUATION_NORTH_GATE, priority: 255, force: ORDERLY"	1KB	<50ms	ADR-026	Signed	Evacuation command
5.4	T+2150ms	Node-5	Node-2	OVERRIDE Command	"EVACUATION_SOUTH_GATE, priority: 255"	1KB	<50ms	ADR-026	Signed	Evacuation command
5.5	T+2150ms	Node-5	Node-3	OVERRIDE Command	"TRANSIT_DIVERSION, priority: 255"	1KB	<50ms	ADR-026	Signed	Diversion command

Step	Timestamp	Origin	Destination	Message Type	Protocol/Payload	Size	Latency	ADR	ZK-Proof	Action
5.6	T+2150ms	Node-5	Node-4	OVERRIDE Command	"VEHICLE_LOCKDOWN, priority: 255"	1KB	<50ms	ADR-026	Signed	Lockdown command
5.7	T+2200ms	Node-1,2,3,4	Node-5	Ack + Status	"Override accepted, executing, ETA: 30s"	500B	<50ms	ADR-007	—	Execution confirmation
5.8	T+2500ms	All Nodes	Node-5 Constellation Hub	Execution Report	"Evacuation 30% complete, no casualties, ZK-Logs attached"	10KB + proofs	100ms	ADR-030	Multiple	Post-action report

SECTION 6: RECOVERY & RECONCILIATION (Disaster Scenario)

Step	Timestamp	Origin	Destination	Message Type	Protocol/Payload	Size	Latency	ADR	ZK-Proof	Action
6.1	T+0 (failure)	Node-2	Node-1, Node-3	SUSPECTED Status	"Node-2 not responding to gossip, T_suspect=2s"	64B	200ms	ADR-004	—	Failure detection
6.2	T+2s	Node-1, Node-3	Node-2	Indirect Probe	"Probe via Node-4: Node-2 reachable?"	128B	200ms	ADR-004	—	Indirect verification
6.3	T+2.5s	Node-4	Node-1, Node-3	Probe Result	"Node-2 FAILED, last seen T-2.5s"	128B	200ms	ADR-004	—	Failure confirmation
6.4	T+3s	Node-3	Node-1, Node-5	Raft Election	"Candidate Node-3, request votes for Zone-A leadership"	1KB	500ms	ADR-004	—	New leader election

Step	Timestamp	Origin	Destination	Message Type	Protocol/Payload	Size	Latency	ADR	ZK-Proof	Action
6.5	T+3.5s	Node-1, Node-5	Node-3	Vote Grant	"Vote granted to Node-3 for Zone-A"	500B	500ms	ADR-004	—	Consensus reached
6.6	T+4s	Node-3	All Zone-A Micro Agents	Leadership Announce	"New leader: Node-3, migrate state"	2KB	100ms	ADR-004	Signed	Leadership assumption
6.7	T+5s	Node-3	Node-5	State Sync Request	"Request last valid state from Node-2 (backup)"	5KB	200ms	ADR-005	—	State recovery
6.8	T+6s	Node-5	Node-3	State Snapshot	"Node-2 state T-2.5s: 184729 events, Merkle: 0xDEF456"	50KB	200ms	ADR-005, ADR-011	Merkle proof	State restore
6.9	T+8s	Node-3	Node-5	Recovery Complete	"Zone-A coverage restored, 80% capacity, ZK-Log attached"	10KB	—	ADR-007	—	Recovery completed

 REAL-TIME PERFORMANCE METRICS SUMMARY

Category	Metric	Node Source	Proof Type	Privacy Level
Safety	Riot/Anomaly Detected	Node 2, 3	Groth16_Anomaly_v2	Critical (Alert only, no video)
Crowd	Density People/m²	Node 2, 3	Groth16_CrowdCount_v1	High (Anonymous number)
Security	Unauthorized Access	Node 4	Groth16_Identity_ZK	Maximum (No ID revealed)
Performance	Avg Inference Latency	All	Signed_Metric_Report	Public (System telemetry)

Category	Metric	Node Source	Proof Type	Privacy Level
Compliance	GDPR Data Retention Check	Node 1	Groth16_Compliance_v1	Audit (Proof of deletion)

 PROTOCOL DETAILS: GOSSIP SWIM

Gossip: Crowd Density Hash Propagation **Protocol:** ADR-004 Gossip SWIM (T_gossip=200ms, k=3) **Payload:** ~500 bytes every 200ms

Message Structure:

yaml

header:

msg_type: DENSITY_UPDATE

origin_node: 0xM-Node-2

sequence: 184729

timestamp: 1709043200

payload:

zone_density_map: {"7A": 4.2, "7B": 3.1, "8A": 2.8,...}

flow_vectors: [dx, dy] per zone

prediction_30s: "CONVERGENCE_TOWARD_GATE_3"

verification:

zk_digest: sha256(zk_proof_of_density_calc)

signature: Ed25519(node_2_key)

merkle_root: root_of_last_100_samples

Convergence: $O(\log N) = 13$ hops per 10K nodes **Latency:** < 2.6s for full mesh propagation

 OPERATIONAL SUMMARY

This 5-node constellation demonstrates how OxMeridian transforms video surveillance:

From (Traditional Cloud)	To (OxMeridian Edge Constellation)
Centralized Processing (All video to cloud)	Distributed Intelligence (Processing at source)
High Bandwidth (20Mbps × Cameras)	Low Bandwidth (192B Proofs + Metadata)
Trust-Based (Trust the provider)	Verify-Based (Mathematical Proof)
Privacy Risk (PII leaves the venue)	Privacy-Preserving (PII stays on node)
Single Point of Failure (Cloud outage)	Resilient Mesh (Self-healing in <5s)
Human Monitoring (Slow reaction)	AI + Human Override (Fast AI, Controlled Human)

REGULATORY COMPLIANCE

- **GDPR:** ZK-Proof guarantees that no PII data leaves the origin node.
- **EU AI Act (Human Oversight):** ADR-026 Cryptographic Override enables effective human intervention & control (Art. 14 §4)
- **ISO 27001:** Immutable cryptographic audit trail (Merkle + ZK).

"The Constellation does not ask for trust. It asks for verification." — ADR-030, OxMeridian Federated Constellation Governance

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